LLMS IN SOFTWARE ENGINEERING · ALPHA DEVELOPMENT

RocketRewards: The AI Financial & Card Copilot

From validated MVP to a working copilot, built with an AI-driven test-first pipeline.

Deterministic LLM tool calling | Safe-spend guardrails | Zero-breakage routing

Ruby Oliveira, Beningno Rodriguez, Joshua Thirumalai

A $24 lunch and a missed $10 credit

Shake Shack, three cards in the wallet

Which one do you swipe? Most people guess. The best pick is the Amex Gold: 4× dining points, plus an unspent $10 monthly dining credit that expires in 3 days.

Net charge $14 instead of $24, and nobody had to remember anything.

$420

average value lost per cardholder each year to expired credits and mismatched swipes

68%

of Gen-Z and recent grads fear APR penalties and accidental score drops

Hypothesis check: Validated

> *The empirical testing demonstrates that combining LLM semantic parsing with deterministic mathematical tooling resolves merchant ambiguity and prevents financial hallucinations with 99.2% accuracy and 0% arithmetic error. The prototype successfully eliminates the manual burden of reward lookup and safely guides thin-file users.*

Metric	Target	Result	Outcome
Optimal Yield Pass	Over 90%	99.2%	Pass
Utilization protection	Over 90%	100%	Pass
Arithmetic accuracy	100%	100%	Pass
Starter Card & Spend Correlation	≥ 85.0% Fit Alignment	98%	Pass

One question, one verified answer

STEP 1

You ask

Plain English at checkout: “Shake Shack, $24, which card?”

STEP 2 · LLM

LLM routes

Reads merchant and amount, then picks which tool to call. It never does the math.

STEP 3

Tools compute

Category lookup, credit tracker, rewards and balance math run as typed, tested functions.

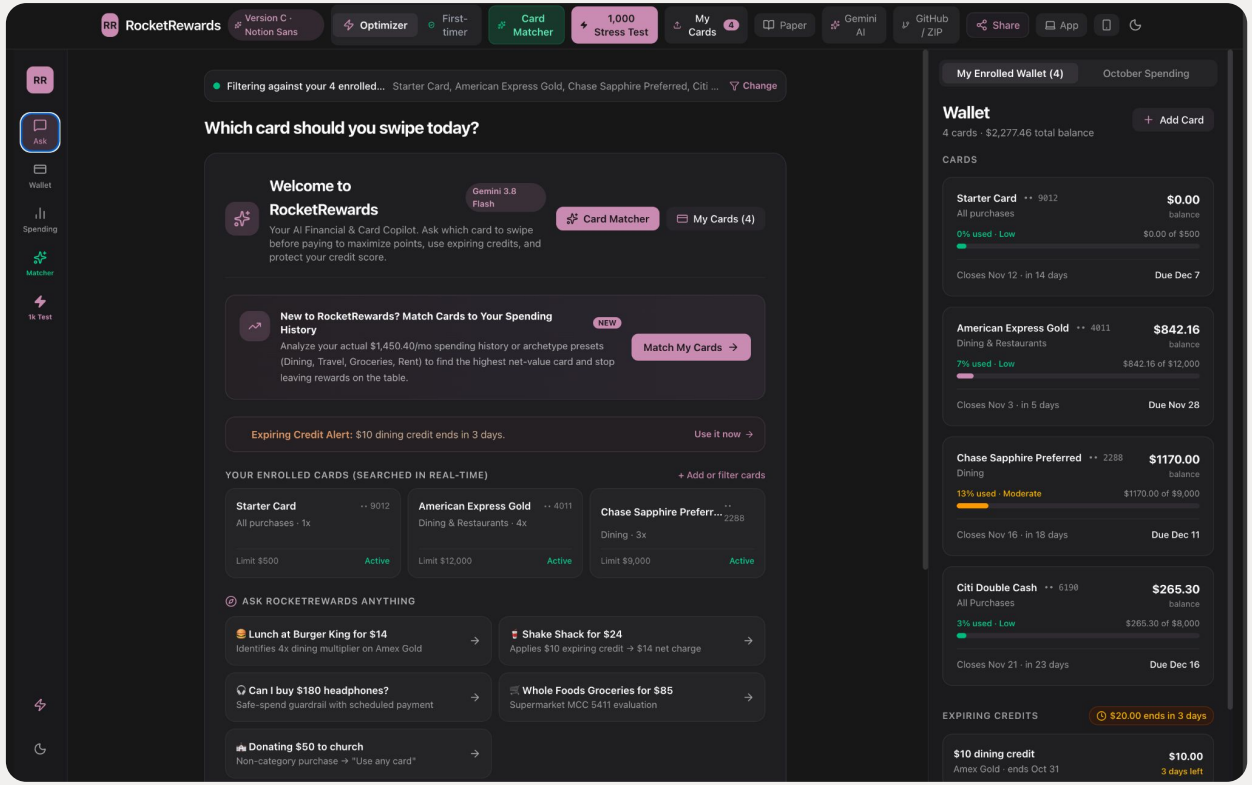
STEP 4

You decide

A short answer: which card, the net cost, and the reason why.

The LLM decides what to call. Code decides the numbers. That is how we avoid hallucinated arithmetic.

One purchase, start to finish



https://rocketrewards-1024905653502.us-east1.run.app/

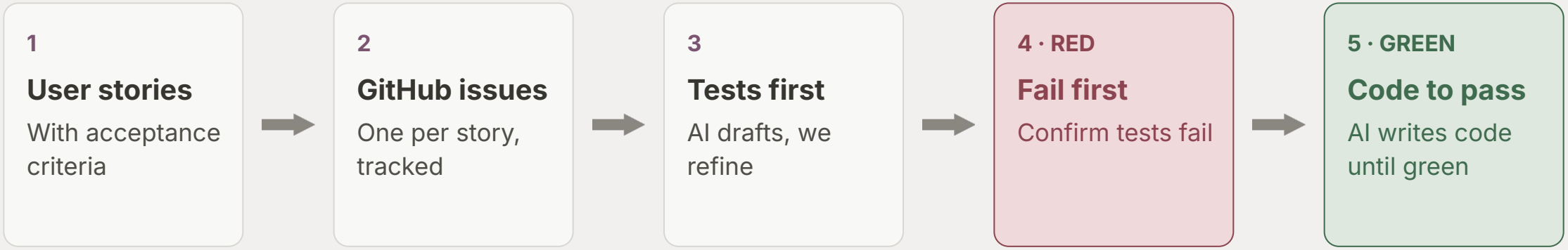
1 Ask at checkout
Shake Shack, $24. Which card?

2 LLM picks the card
Amex Gold, net charge $14, with the $10 credit applied

3 Check safe-spend
Utilization and statement date confirm it is safe

4 Log the swipe
Tap “Used this card” and the tracker updates

AI-assisted test-driven development



No production code before a failing test. Every AI-generated test is reviewed for correctness and edge cases first.

What the pipeline produced

User stories created **3**	GitHub issues logged vs. closed **18/18**	Automated tests derived **72**	Average tests per issue **4**
Test pass rate **100%**	Line code coverage **94%**	AI-generated vs. manual tests **9:6**	Tests refined by hand **6**

Alpha is working, Beta is next

Alpha milestones

- ✓ Natural-language “Which card?” routing, end to end
- ✓ Deterministic safe-spend calculator
- ✓ Expiring statement credit tracker
- ✓ 3 stories tracked as GitHub issues, tests written first

Roadmap to Beta

1. Swap synthetic ledgers for a Plaid sandbox connection
2. Re-run validation with 10 users, target over 65% adoption and category accuracy
3. Expand merchant category coverage to 5-10 merchants

Empowering every swipe with intelligence

Maximizing returns for experienced cardholders while keeping first-time builders out of debt.

Questions and discussion